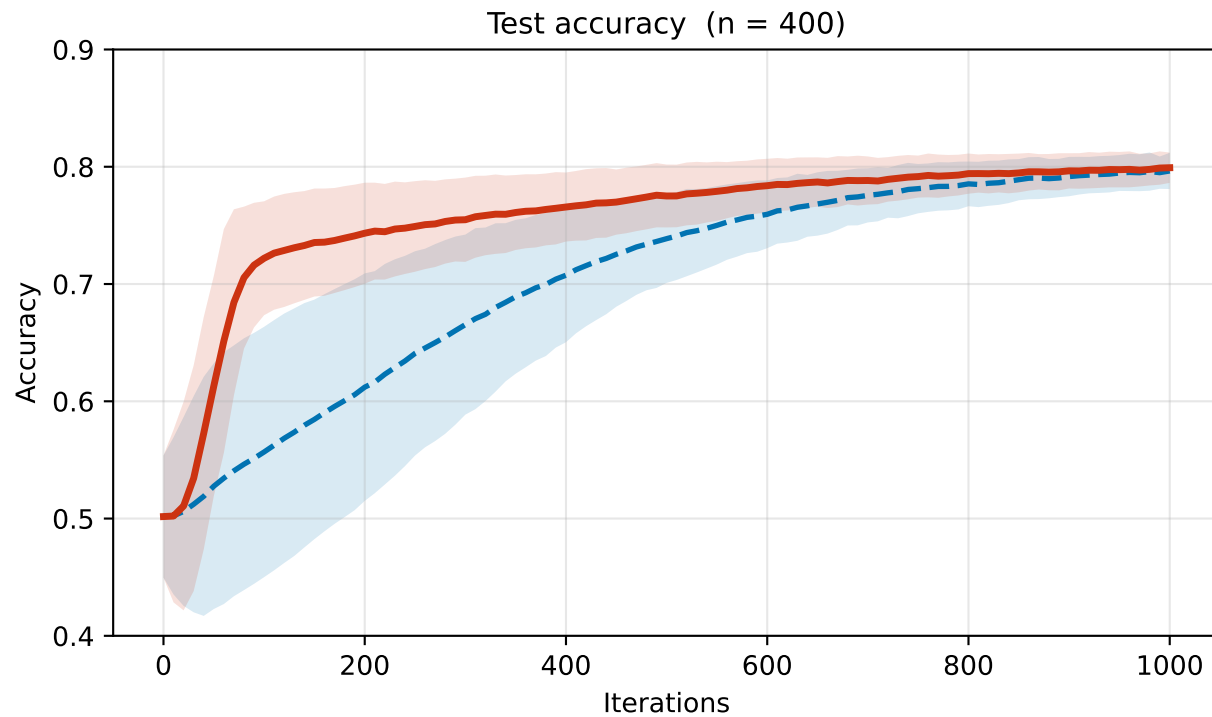
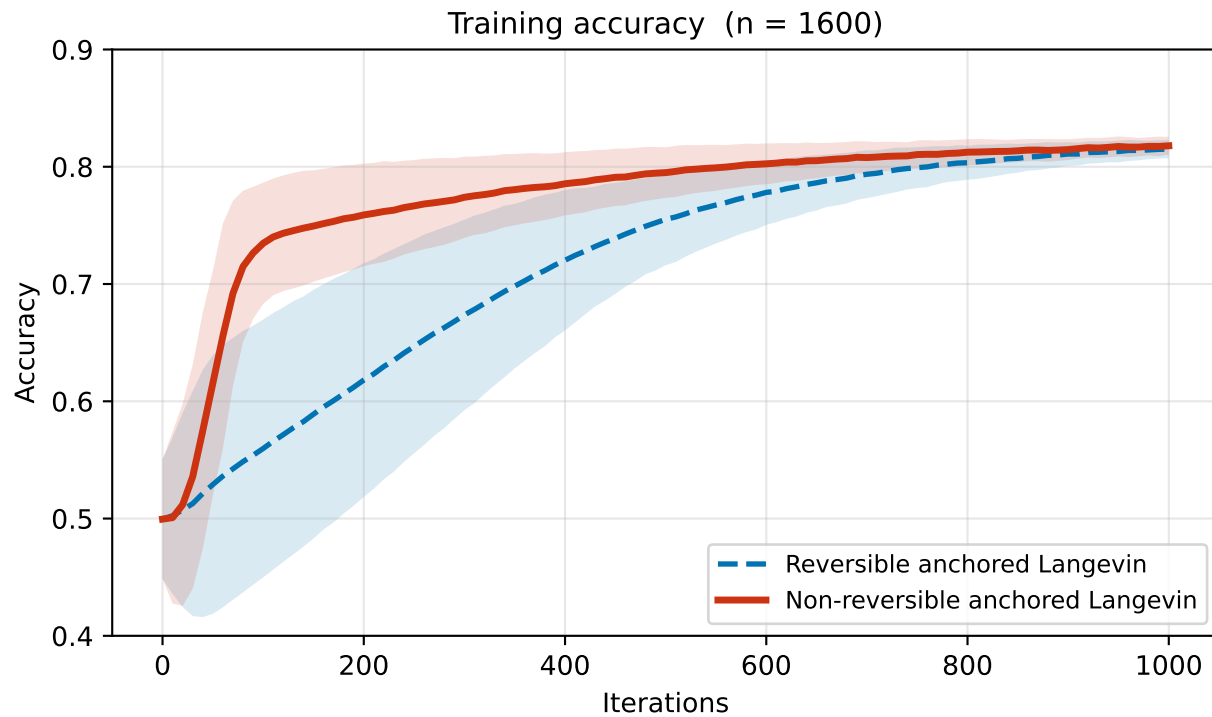


Non-reversible beats reversible anchored Langevin — L1-smooth ball



$d = 9$ (intercept + 8 slopes); constraint: L1-smooth ball; $U = f + g$ with $g = 2 \cdot \sum |w_j|$ (non-differentiable), anchor $\delta = 0.02$; EXACT gradient; $\eta = 7.0e-06$; $s = 4$; $R = 100$; $\alpha = 0$ vs 1.

Design: unstandardised covariates -- in each slope triple the covariance is $1 q_1 q_1' + 64 q_2 q_2' + 2 q_3 q_3'$ with $q_1 = (1, 1, 1)/\sqrt{3}$ (the rotation axis), $q_2 = (0, 1, -1)/\sqrt{2}$ (fast, no signal), $q_3 = (2, -1, -1)/\sqrt{6}$ (slow, in the rotated plane, carries signal); $\beta_{\text{triple}} = (1.5, 0.25, 0.25)$.

Lines: across-replicate mean; bands: mean ± 1 sample sd of single-iterate accuracy (repeat-run variability, not confidence intervals). Shared initialisation and noise per replicate. y-axis fixed to (0.4, 0.9) -- NOT windowed on the plateau.